CSD 208

Assignment

Comparison of State-of-the-Art Machines

Characteristic	November '18	June '18	November '17	June '17	November '16	June '16	November '15	June '15	November '14	June '14
#1 Machine	**Summit**	**Summit**	**Sunway TaihuLight**	**Sunway TaihuLight**	**Sunway TaihuLight**	**Sunway TaihuLight**	**Tianhe-2A**	**Tianhe-2A**	**Tianhe-2A**	**Tianhe-2A**
GENERAL										
Country of manufacture	USA 🇺🇸	USA 🇺🇸	China 🇨🇳	China 🇨🇳	China 🇨🇳	China 🇨🇳	China 🇨🇳	China 🇨🇳	China 🇨🇳	China 🇨🇳
Site	DOE/SC/Oak Ridge National Laboratory	DOE/SC/Oak Ridge National Laboratory	National Supercomputing Center in Wuxi	National Supercomputing Center in Wuxi	National Supercomputing Center in Wuxi	National Supercomputing Center in Wuxi	National Super Computer Center in Guangzhou	National Super Computer Center in Guangzhou	National Super Computer Center in Guangzhou	National Super Computer Center in Guangzhou
Manufacturer	IBM	IBM	NRCPC	NRCPC	NRCPC	NRCPC	NUDT	NUDT	NUDT	NUDT
# of cores	23,97,824	23,97,824	1,06,49,600	1,06,49,600	1,06,49,600	1,06,49,600	49,81,760	49,81,760	49,81,760	49,81,760

Processor	IBM POWER9 22C 3.07GHz	IBM POWER9 22C 3.07GHz	Sunway SW26010 260C 1.45 GHz	Sunway SW26010 260C 1.45 GHz	Sunway SW26010 260C 1.45 GHz	Sunway SW26010 260C 1.45 GHz	Intel Xeon E5-2692v2 12C 2.2GHz	Intel Xeon E5-2692v2 12C 2.2GHz	Intel Xeon E5-2692v2 12C 2.2GHz	Intel Xeon E5-2692v2 12C 2.2GHz
Interconnect	Dual-rail Mellanox EDR Infiniband	Dual-rail Mellanox EDR Infiniband	Sunway	Sunway	Sunway	Sunway	TH Express-2	TH Express-2	TH Express-2	TH Express-2
Memory	2,801,664 GB	2,801,664 GB	1,310,720 GB	1,310,720 GB	1,310,720 GB	1,310,720 GB	2,277,376 GB	2,277,376 GB	2,277,376 GB	2,277,376 GB
PERFORMANCE										
Linpack Performance [Rmax]	143,500 TFlop/s	122,300 TFlop/s	93,014.6 TFlop/s	93,014.6 TFlop/s	93,014.6 TFlop/s	93,014.6 TFlop/s	61,444.5 TFlop/s	61,444.5 TFlop/s	61,444.5 TFlop/s	61,444.5 TFlop/s
Theoretical Peak [Rpeak]	200,795 TFlop/s	200,795 TFlop/s	125,436 TFlop/s	125,436 TFlop/s	125,436 TFlop/s	125,436 TFlop/s	100,679 TFlop/s	100,679 TFlop/s	100,679 TFlop/s	100,679 TFlop/s
Nmax	16693248	16693248	12288000	12288000	12288000	12288000	9773000	9773000	9773000	9773000
POWER CONSUMPTION										
Power	9,783.00 kW	9,783.00 kW	15,371.00 kW	15,371.00 kW	15,371.00 kW	15,371.00 kW	18,482.00 kW	18,482.00 kW	18,482.00 kW	18,482.00 kW
Power Measurement Level	3	3	2	2	2	2	1	1	1	1
SOFTWARE										
Operating System	RHEL 7.4	RHEL 7.4	Sunway RaiseOS 2.0.5	Sunway RaiseOS 2.0.5	Sunway RaiseOS 2.0.5	Sunway RaiseOS 2.0.5	Kylin Linux	Kylin Linux	Kylin Linux	Kylin Linux

Observations

- This table is an interesting analogy for life. One can never really stay at the top long enough. Better and faster competitors always do show up. Take the example of Chinese Tianhe-2A, which stayed at the top spot for six lists, i.e., 3 years. It now trails at fourth spot and has less than half the HPL performance of top-rated Summit.

- The past year has proved to be a good one for United States in this everlasting war for supercomputer supremacy. China has been displaced from the top seat it had held for so many years and the United States is firmly in the driver's seat with two of its machines at the top. One might just wonder how far behind India might be in this race, but then again, the more we think about that, the worse we'll feel. I wonder what way have happened, have the Chinese become relaxed and complacent or did the US up its game far too much? Anyhow, the next few lists sure promise to be interesting. The battle is on.

- One is really keen to see where Sierra goes on the next list, given its mammoth jump from 71.6 petaflops to 94.6 petaflops from Jun 18 to Nov 18, which helped it overtake the quadruple list leader, Sunway TaihuLight. It may be interesting to note that both Summit and Sierra are IBM built supercomputers, powered by Power9 CPUs and NVIDIA V100 GPUs. It is similar to Formula1 where a team often has two racing greats, who compete against each other but are also in the same team.

- The difference between the 1st and the 10th computers is just astounding, with the performance of the Sequoia being just about 12% of the Summit. It is interesting to note that Sequoia, which we just wrote off, was nothing short of a powerhouse and has stayed at the 3rd position for a long time as recently as 3 years ago. This just goes to show the massive rate at which improvements are being made in this field.It is surprising to note that in almost all the lists, the top few supercomputers are not very close together in terms of HPL performance, which is contrary to what one would normally expect in any sort of a leaderboard. That is probably the reason that there is usually no competition between existing supercomputers. It is rare to see two machines going neck to neck. The general trend has been that a machine dominates the board for a few years and is then succeeded by a machine that completely overpowers it, and the same happends again and again, which is why we see the huge difference and hardly any recurring competition.

- In the Nov 2018 list, it is funny to see that Sunway TaihuLight has a much higher number of cores and power consumption but a much lower performance benchmark than Summit.

- We see that China has really upped its game, as prior to June 2016, it had never been the case that the U.S. was not home to the largest number of TOP500 computers.

- It is kind of surprising to see a Saudi Arabian system, Shaheen II on some of the lists, and it just seems like an investment of money done to get on the list.

- What we see in the past two lists is an effort by the US Summit to cement its position at the top. It's the only supercomputer for the past five years that has upgraded its performance despite already having the top spot, which is something that the Chinese computers never did. Their HPL mark stayed constant year after year. Even the US Sierra has made massive leaps, which just goes to show the intent of the US in this game. This is probably in response to the fact that in the November 2017 list, China claimed 202 out of 500 computers while US had just 143, which were records on either end, and is an embarrassingly different situation from Nov 2015, when US had 200 and China had 108. This reminds me of the Us-USSR space race, and just goes to show how much the US loves races

- We are once again reminded of the sheer power of these two nations that rule the roost when it comes to supercomputers. IN the past 5 years (10 lists), the only non-US or Chinese machines are – JUQUEEN (Germany), Piz Daint (Switzerland), K Computer (Japan), Shaheen II (Saudi Arabia), Hazel Hen (Gerrmany), Oakforest-PACS (Japan), Gyoukou (Japan), AI Bridging Cloud Infrastructure (Japan), SuperMUC-NG (Germany), and these have hardly ever made it to the top three, revealing the cutthroat competition at the top.

- Given all the improvements, it is surprising to see that the top performer from Jun 2014, if present in the current list, would still have made it at 4th position. This ince again show the lack of frequent shuffling at the top and the only replacement happening when a new computer completely outperforms the previous topper. We see Summit trying to avoid such a situation and really boosting its performance to prevent such a situation from happening to it.

- Like USA and China, HP and IBM also continue to battle for supercomputer supremacy and continue a tough neck-to-neck battle.

- Nearly all the systems have 6 or more cores while about 85% have 8 or more cores.